

Partitions into Catalan Numbers

All-Orders Asymptotics, Periodic Corrections, and Inverse Growth

A proof-focused research report for the ProveIt project

October 1, 2026

Abstract

Let $p(n)$ count ordinary integer partitions whose allowed part sizes are $1, 2, 5, 14, 42, \dots$, the distinct Catalan numbers. This is OEIS A033552. We develop a complete expansion in inverse powers of a logarithmic size parameter, including a nonconstant periodic amplitude, an explicit first correction, and an inverse expansion for the threshold at which $p(n)$ reaches a prescribed value. The argument combines an all-orders coefficient saddle estimate, a Barnes G product identity, and an exponentially convergent Fourier series involving $\Gamma(s)\zeta(1+s)$. The contour estimate controls all noncentral arcs; the expansion is therefore more than a formal saddle calculation. A further theorem gives a phase-dependent limiting distribution for the largest allowed-part index in a uniformly chosen partition. Exact computations through $n = 10^6$ and independent high-precision checks accompany the proofs. The inspected OEIS entry contains no asymptotic formula; we do not claim that it records an explicit asymptotic conjecture, or that an exhaustive literature-priority search has been completed. The sequence-specific formulas and proofs below are the substantive output, not a claim that the classical saddle or periodic-partition methods themselves are new.

Principal formula. Put $\alpha = \log 4$ and let μ be the large solution of $n = 4^\mu / \sqrt{\pi\mu}$. Then

$$p(n) = e^{C + \Psi_\alpha(\mu)} \mu^{23/8} \exp\left(\frac{\alpha}{2}\mu^2 - \frac{1 + 3\alpha}{2}\mu\right) \left(1 + \frac{D_\alpha(\mu)}{\mu} + O(\mu^{-2})\right).$$

Here C is explicit and Ψ_α, D_α are smooth, one-periodic functions given below. Every subsequent inverse-power coefficient is constructively determined. Replacing Ψ_α by a constant does *not* give an asymptotic equivalent.

[Write note, batch 7301, 1 October 2026.] This is the collection edition of the report, written into ProveIt's research-report collection from manuscript 39 of batch 7301 (provenance in [appendix C](#)). Every label carries the prefix `ctp::`; `requirements.txt` now lives in `data/` and `build.sh` in `code/`. Text added in the write is marked like this note; everything else is the delivered text. As the abstract says, the OEIS entry states *no* asymptotic conjecture: this report answers no posed question, it supplies the asymptotics the entry lacks.

Contents

1	The problem, its status, and the contribution	3
1.1	The sequence and the normalization	3
1.2	What is established here, and what is not	3
1.3	Relation to the ProveIt repository	4
2	Main asymptotic and inverse formulas	4
2.1	Constants and the periodic amplitude	4
2.2	The inverse threshold	6
3	An all-orders coefficient saddle estimate	6
3.1	Positive cumulants and the saddle	6
3.2	The noncentral arcs: why the coefficient estimate is valid	7
3.3	The central expansion and its remainder	8
3.4	A reusable sufficient hypothesis	9
4	The Catalan product and its moving periodic phase	9
4.1	The logarithm of a Catalan number and the Barnes product	9
4.2	Separating the bulk from a localized lattice sum	9
4.3	The first two bulk orders, with a uniform remainder	11
4.4	A constructive rule for every coefficient	11
5	Deriving the explicit coefficient expansion	12
5.1	The saddle coordinate and its correction	12
5.2	The normalization and the first correction	13
5.3	Completion of the all-orders proof	13
6	Inverse growth and the integer staircase	14
6.1	Reversion of the quadratic logarithmic core	14
6.2	Why a continuous calculation describes the discrete threshold	14
6.3	A practical caution	14
7	The largest Catalan part in a random partition	15
8	A universality statement for polynomially perturbed geometric parts	16
9	Computation, independent checks, and reproducibility	17
9.1	What the supplied programs do	17
9.2	Coefficient accuracy	17
9.3	Inverse and largest-index checks	18
9.4	Numerical safeguards and build instructions	18
10	Further research questions and concrete next steps	19
10.1	Certified errors and exact inverse decisions	19
10.2	A symbolic compiler for periodic asymptotic coefficients	19
10.3	Beyond-all-orders contributions	20
10.4	Joint statistics near the largest part	20
10.5	The number of parts and mixed conditioning	20
10.6	Fuss–Catalan and other gamma-quotient part sequences	20
10.7	Exact counting complexity	20
10.8	Formalization and a periodic coefficient algebra	20
11	Conclusion	21
A	A compact audit of the algebra	21
B	Provenance and scope of the search	21
C	Provenance of the collection edition	22

1 The problem, its status, and the contribution

1.1 The sequence and the normalization

For $k \geq 1$ define

$$c_k = \frac{1}{k+1} \binom{2k}{k}, \quad F(z) = \prod_{k \geq 1} (1 - z^{c_k})^{-1}, \quad p(n) = [z^n] F(z). \quad (1.1)$$

Thus the allowed part sizes are $c_1 = 1, c_2 = 2, c_3 = 5, c_4 = 14, \dots$. The initial terms are

$$1, 1, 2, 2, 3, 4, 5, 6, 7, 8, 10, 11, 13, 14, 17, 19, 22, 24, 27, 30, \dots$$

The index starts at $n = 0$. It is important not to include both c_0 and c_1 : doing so would create two colors of the unit part and a different sequence. The definition and generating function agree with OEIS A033552 [1].

The inspected entry supplies a generating function, a recursive computation, and a reference to Pak's enumerative-complexity survey, but no asymptotic formula. Pak explicitly discusses partitions into Catalan numbers among restricted-partition counting problems that may be less tractable than some classical examples [2, §2.5]. That discussion concerns exact counting complexity. An asymptotic expansion, however accurate to fixed order, does not by itself settle that complexity question.

All logarithms in this article are natural. A periodic function is evaluated on a real argument by periodic extension. All O -estimates concern the limit of that argument, or of n , to positive infinity; their constants may depend on a fixed truncation order but not on a fractional phase.

1.2 What is established here, and what is not

The principal contribution is a self-contained chain from the exact product (1.1) to the following results: an all-orders saddle expansion with a contour proof; a completely normalized, phase-resolved explicit expansion; a first correction containing derivatives of the periodic amplitude; an inverse threshold expansion with error transport; and a phase-dependent law for the largest part index. A general proposition explains which coefficients depend only on the exponential and polynomial growth of the allowed part sizes.

Periodicity in asymptotics of geometrically restricted partitions is a classical phenomenon. Erdős and Richmond discuss the de Bruijn–Mahler phenomenon and its extensions [3]. Mellin analysis of harmonic sums is developed systematically by Flajolet, Gourdon, and Dumas [4]. We use that context, but prove the concrete estimates and normalizations needed here rather than importing a vaguely matched theorem. The explicit Catalan-part formulas below were derived for this report. A limited search found no source giving this complete collection of formulas; that observation is not a proof of publication priority.

[Write note, batch 7301, 1 October 2026.] The phenomenon Erdős and Richmond name goes back to de Bruijn's treatment of Mahler's partition problem [9], which the delivered bibliography did not list: partitions into powers of a fixed integer, whose logarithm carries a periodic function of a logarithmic variable built from $\Gamma(s)\zeta(1+s)$ on the Mellin lattice. The Catalan parts $c_k \sim 4^k/(\sqrt{\pi}k^{3/2})$ are a polynomially perturbed geometric sequence of that kind, and lemma 4.1 is the corresponding Mellin computation; see also the note after its proof.

The manuscript contains conventional mathematical proofs and executable checks. It is not a Lean formalization. The numerical checks corroborate, but do not replace, the arguments. No computer search is used as a substitute for a missing contour estimate or for a missing asymptotic remainder.

1.3 Relation to the ProveIt repository

The inspected repository snapshot is commit `e6190a94552f119e9a2140e5fea8450a045a0837`. Its transseries documentation separates asymptotic scales from inversion, distinguishes a continuous inverse from an integer staircase inverse, and records qualifications on formalized and unformalized results [7]. Those distinctions are useful here. The companion discusses asymptotics of Catalan numbers themselves, whereas our object is the partition sequence whose *allowed parts* are Catalan numbers. An initial repository code search for A033552 returned no match; that is a duplication check, not a mathematical premise.

The present article uses an exactly invertible dominant size coordinate and then transports residual errors through inversion. Its periodic coefficients also make a useful boundary case for a purely polynomial–logarithmic transseries calculus: the coefficient algebra must retain a periodic phase.

[Write note, batch 7301, 1 October 2026.] At the repository’s current state the search result above still holds: no report or formal development treats A033552 or partitions into Catalan numbers. The pin quoted above is a commit; the delivered `SOURCE_NOTES.md` also names `1f1981f682b2878bde51a6ad40c22777f362fc05`, the commit its first repository request returned, and calls it “a tree SHA, not a commit SHA”. That is mistaken: it is a ProveIt commit of 1 October 2026 (the root tree of `e6190a945` is the separate object `8e77886c`). Neighbouring material is listed in [appendix C](#). No statement of this report is formalized.

2 Main asymptotic and inverse formulas

2.1 Constants and the periodic amplitude

Set

$$\alpha = \log 4, \quad A = \frac{1+3\alpha}{2}, \quad \eta = \frac{23}{8}, \quad C = \frac{1}{4} \log 2 + \frac{3}{4} \log \pi + \log G(3/2), \quad (2.1)$$

where G is the Barnes function, normalized by $G(1) = 1$ and $G(z+1) = \Gamma(z)G(z)$ [5]. For the moment allow $\alpha > 0$ to vary, and define

$$\kappa = \frac{\pi^2}{12} - \frac{\gamma^2}{2} - \gamma_1, \quad (2.2)$$

$$\Psi_\alpha(\theta) = \frac{\alpha}{12} + \frac{\kappa}{\alpha} + \frac{1}{\alpha} \sum_{\ell \in \mathbb{Z} \setminus \{0\}} \Gamma\left(\frac{2\pi i \ell}{\alpha}\right) \zeta\left(1 + \frac{2\pi i \ell}{\alpha}\right) e^{2\pi i \ell \theta}. \quad (2.3)$$

Here γ is Euler’s constant and γ_1 is the first Stieltjes constant, with convention $\zeta(1+s) = s^{-1} + \gamma - \gamma_1 s + O(s^2)$. The Fourier series and all of its derivatives converge absolutely. Conjugate terms pair to make Ψ_α real. A prime denotes differentiation with respect to θ ; a partial α -derivative holds θ fixed. The correction is

$$D_\alpha(\theta) = \frac{1}{6} - \frac{1}{2\alpha} - \frac{3}{2} \partial_\alpha \Psi_\alpha(\theta) + \frac{17}{8\alpha} \Psi'_\alpha(\theta) - \frac{\Psi'_\alpha(\theta)^2 + \Psi''_\alpha(\theta)}{2\alpha^2}. \quad (2.4)$$

Only after taking the partial derivative do we set $\alpha = \log 4$.

Define the large real size coordinate

$$n = \frac{4^{\mu(n)}}{\sqrt{\pi \mu(n)}}, \quad \mu(n) = -\frac{1}{2\alpha} W_{-1}\left(-\frac{2\alpha}{\pi n^2}\right). \quad (2.5)$$

The negative real branch W_{-1} is intended. For sufficiently large n this is the unique large solution and $\mu(n) \sim (\log n)/\alpha$.

Theorem 2.1 (Explicit expansion, including the first correction). *For the sequence (1.1), as $n \rightarrow \infty$ through integers,*

$$\log p(n) = \frac{\alpha}{2}\mu^2 - A\mu + \eta \log \mu + C + \Psi_\alpha(\mu) + \frac{D_\alpha(\mu)}{\mu} + O(\mu^{-2}). \quad (2.6)$$

Equivalently,

$$p(n) = e^{C+\Psi_\alpha(\mu)} \mu^\eta e^{\alpha\mu^2/2-A\mu} \left(1 + \frac{D_\alpha(\mu)}{\mu} + O(\mu^{-2})\right). \quad (2.7)$$

For every fixed $J \geq 0$ there are smooth, one-periodic functions $L_1, \dots, L_J$, with $L_1 = D_\alpha$, such that

$$\log p(n) = \frac{\alpha}{2}\mu^2 - A\mu + \eta \log \mu + C + \Psi_\alpha(\mu) + \sum_{j=1}^J \frac{L_j(\mu)}{\mu^j} + O_J(\mu^{-J-1}). \quad (2.8)$$

Sections 3–5 prove this statement and give a finite coefficient-construction procedure at each order.

The phrase “all orders” here means a remainder theorem for every fixed truncation order. It does not assert convergence of the resulting infinite inverse-power series, nor does it resolve exponentially smaller sectors.

Corollary 2.2 (An unavoidable periodic factor). *There is no constant B for which*

$$p(n) \sim B\mu(n)^\eta e^{\alpha\mu(n)^2/2-A\mu(n)}.$$

More precisely, the limit points of the normalized sequence are exactly

$$\left[\exp\{C + \min_\theta \Psi_\alpha(\theta)\}, \exp\{C + \max_\theta \Psi_\alpha(\theta)\} \right],$$

an interval with distinct endpoints.

Proof. The gamma function has no zeros, and $\zeta(1+it) \neq 0$ for real $t \neq 0$ [6]; hence the first Fourier coefficient in (2.3) is nonzero. Thus Ψ_α is nonconstant. For any fixed phase θ , round $4^{j+\theta}/\sqrt{\pi(j+\theta)}$ to the nearest integer n_j . Implicit differentiation of (2.5) shows $\mu(n_j) = j + \theta + o(1)$. Every phase therefore occurs as a subsequential limit. Apply (2.7) and continuity of Ψ_α . $\square$

The first Fourier harmonic has amplitude approximately 0.00100279457386. The modulation is small but does not tend to zero. Its mean is approximately 0.64116608028709. These are reproducible floating-point evaluations, not interval-certified bounds.

Corollary 2.3 (The first three logarithmic growth terms). *With $X = \log n$,*

$$\log p(n) = \frac{X^2}{2\alpha} + \frac{X \log X}{2\alpha} + \left(\frac{\log(\pi/\alpha) - 1}{2\alpha} - \frac{3}{2} \right) X + O((\log X)^2). \quad (2.9)$$

Proof. The defining equation is $X = \alpha\mu - \frac{1}{2} \log \mu - \frac{1}{2} \log \pi$. Hence

$$\alpha\mu = X + \frac{1}{2} \log X + \frac{1}{2} \log(\pi/\alpha) + O((\log X)/X).$$

Substitute this into (2.6) and collect terms of orders X^2 , $X \log X$, and X . $\square$

The positive sign of $X \log X$ is a concrete effect of the factor $k^{-3/2}$ in the growth of c_k . Replacing c_k by 4^k too early loses that information.

2.2 The inverse threshold

For real $y > 1$ define the integer staircase

$$N(y) = \min\{n \geq 0 : p(n) \geq y\}. \quad (2.10)$$

Put

$$Y = \log y, \quad R = \sqrt{2\alpha Y}, \quad s = R/\alpha. \quad (2.11)$$

Theorem 2.4 (Inverse growth and its first correction). *As $y \rightarrow \infty$,*

$$\begin{aligned} \log N(y) = R + A - \frac{1}{2} \log s - \frac{1}{2} \log \pi \\ + \frac{\frac{1}{2}(A^2 - A) - \alpha \eta \log s - \alpha \{C + \Psi_\alpha(s + A/\alpha)\}}{R} + O\left(\frac{(\log R)^2}{R^2}\right). \end{aligned} \quad (2.12)$$

In particular,

$$N(y) \sim 8 \sqrt{\frac{e\alpha}{\pi}} \frac{\exp(\sqrt{2\alpha \log y})}{(2\alpha \log y)^{1/4}}. \quad (2.13)$$

The inverse has an expansion to arbitrary fixed order in R^{-1} , with coefficients polynomial in $\log R$ and smooth periodic functions of $s + A/\alpha$. Section 6 proves the passage from a smooth asymptotic inverse to the integer threshold.

3 An all-orders coefficient saddle estimate

3.1 Positive cumulants and the saddle

For $t > 0$ write

$$H(t) = \log F(e^{-t}), \quad K_j(t) = (-1)^j H^{(j)}(t) = \sum_{k \geq 1} c_k^j \sum_{\ell \geq 1} \ell^{j-1} e^{-\ell t c_k} \quad (j \geq 1). \quad (3.1)$$

Absolute convergence on each half-plane $\operatorname{Re} t \geq t_0 > 0$ justifies termwise differentiation. Since $K'_1(t) = -K_2(t) < 0$, K_1 decreases strictly from infinity to zero. Thus, for each $n > 0$, there is a unique $t_n > 0$ satisfying

$$K_1(t_n) = n. \quad (3.2)$$

The real logarithmic interpolation of the part sizes is

$$r(x) = \alpha x - \frac{1}{2} \log \pi + \log \Gamma(x + \frac{1}{2}) - \log \Gamma(x + 2), \quad x \geq 1. \quad (3.3)$$

It obeys $r(k) = \log c_k$. Since $r'(1) = 1/2$ and $r''(x) = \psi'(x + 1/2) - \psi'(x + 2) > 0$, it is strictly increasing, with $r'(x) \rightarrow \alpha$. Define

$$u_n = \log(1/t_n), \quad m_n = r^{-1}(u_n). \quad (3.4)$$

The parameters m_n and $\mu(n)$ are not identical; their small difference is responsible for several cancellations in the explicit formula.

Lemma 3.1 (Size of the scaled cumulants). *Let $t = e^{-r(m)}$ and $m \rightarrow \infty$. For each fixed integer $j \geq 1$,*

$$B_j(t) := t^j K_j(t) = (j-1)! m + O_j(1). \quad (3.5)$$

In particular $nt_n \sim m_n$ and $m_n \sim (\log n)/\alpha$.

Proof. Set $f_j(v) = v^j \sum_{\ell \geq 1} \ell^{j-1} e^{-\ell v}$. For $0 < v \leq 1$, $f_j(v) = (j-1)! + O_j(v)$, by the elementary Laurent expansion of the geometric sum and its derivatives. For $v \geq 1$, $f_j(v)$ decays exponentially up to a fixed polynomial factor. The exact ratio

$$\frac{c_{k+1}}{c_k} = \frac{2(2k+1)}{k+2} \in [2, 4) \quad (3.6)$$

implies $\sum_{tc_k \leq 1} tc_k = O(1)$ and a bounded sum of the large- v tails. There are exactly $\lfloor m \rfloor$ parts with $tc_k \leq 1$. Summing $f_j(tc_k)$ proves (3.5). At the saddle, $nt = B_1 = m + O(1)$, and $\log n = r(m) + \log(m + O(1)) = \alpha m + O(\log m)$. $\square$

Theorem 3.2 (All-orders saddle expansion). *For every fixed $J \geq 0$, with all cumulants evaluated at $t = t_n$,*

$$p(n) = \frac{\exp(H(t) + nt)}{\sqrt{2\pi K_2(t)}} \left(1 + \sum_{j=1}^J E_j(t) + O_J(m_n^{-J-1}) \right). \quad (3.7)$$

The correction E_j is the finite sum

$$E_j = \sum_{\substack{k_3, \dots, k_{2j+2} \geq 0 \\ \sum_{r=3}^{2j+2} (r-2)k_r = 2j}} (-1)^{R/2} (R-1)!! \prod_{r=3}^{2j+2} \frac{1}{k_r!} \left(\frac{K_r}{r! K_2^{r/2}} \right)^{k_r}, \quad R = \sum_{r=3}^{2j+2} r k_r. \quad (3.8)$$

Here R is even, and $E_j = O_j(m_n^{-j})$. In particular,

$$E_1 = \frac{K_4}{8K_2^2} - \frac{5K_3^2}{24K_2^3}, \quad (3.9)$$

$$E_2 = -\frac{K_6}{48K_2^3} + \frac{7K_3K_5}{48K_2^4} + \frac{35K_4^2}{384K_2^4} - \frac{35K_3^2K_4}{64K_2^5} + \frac{385K_3^4}{1152K_2^6}. \quad (3.10)$$

3.2 The noncentral arcs: why the coefficient estimate is valid

The Cauchy coefficient integral is

$$p(n) = \frac{e^{H(t)+nt}}{2\pi} \int_{-\pi}^{\pi} \exp\{H(t-i\theta) - H(t) - in\theta\} d\theta. \quad (3.11)$$

Its normalized absolute value satisfies the exact identity

$$\left| \frac{F(e^{-t+i\theta})}{F(e^{-t})} \right| = \prod_{k \geq 1} \left(1 + \frac{4e^{-tc_k} \sin^2(c_k \theta/2)}{(1 - e^{-tc_k})^2} \right)^{-1/2}. \quad (3.12)$$

All factors on the right are at most one. Fix a requested error order. In the estimates below, $B > 0$, the number of fixed factors K , and a small $\delta > 0$ are chosen in that order and then held fixed as $m \rightarrow \infty$.

For $|\theta| \leq tm^B$, use every part with $c_k \leq 1/(tm^B)$. There are $m - O(\log m)$ such parts. For them, $|c_k \theta| \leq 1$, and elementary sine and exponential estimates give

$$\left| \frac{F(e^{-t+i\theta})}{F(e^{-t})} \right| \leq \left(1 + c(\theta/t)^2 \right)^{-c'm} \quad (3.13)$$

for fixed positive c, c' . Between $t \log m / \sqrt{m}$ and t , integration of this bound gives $O(t \exp\{-c''(\log m)^2\})$, after changing constants. Between t and tm^B , it gives $O(tm^B e^{-c'm})$. Both are smaller than $(t/\sqrt{m})m^{-L}$ for every fixed L .

Next, for $tm^B \leq |\theta| \leq \delta$, select the first K factors and choose $\delta c_K \leq 1$. Each selected factor in (3.12) is at most $C_K t/|\theta|$. Hence

$$\int_{tm^B \leq |\theta| \leq \delta} \left| \frac{F(e^{-t+i\theta})}{F(e^{-t})} \right| d\theta = O_K \left(t m^{-B(K-1)} \right). \quad (3.14)$$

Taking K sufficiently large makes this negligible to any specified inverse-power order relative to the main width $t/\sqrt{m}$.

Finally consider $\delta \leq |\theta| \leq \pi$. The unit-part factor is $O_\delta(t)$. The functions $\sin(2\theta/2)$ and $\sin(5\theta/2)$ do not vanish simultaneously on this compact set: a simultaneous zero would require $\theta \in 2\pi\mathbb{Z}$ because $\gcd(2, 5) = 1$. Therefore at least one of the part-2 and part-5 factors is uniformly $O_\delta(t)$. Thus

$$\int_{\delta \leq |\theta| \leq \pi} \left| \frac{F(e^{-t+i\theta})}{F(e^{-t})} \right| d\theta = O_\delta(t^2). \quad (3.15)$$

Since $t = \exp(-\alpha m + O(\log m))$, this is again negligible to every inverse power of m relative to $t/\sqrt{m}$.

These estimates eliminate possible competing root-of-unity arcs at the precision claimed in [theorem 3.2](#). They do not assert that all those arcs vanish, or identify their exponentially smaller contributions.

3.3 The central expansion and its remainder

At the saddle the linear term in (3.11) cancels. With $v = \theta\sqrt{K_2}$, the exponent becomes

$$-\frac{v^2}{2} + \sum_{r \geq 3} \frac{i^r K_r}{r! K_2^{r/2}} v^r. \quad (3.16)$$

For each fixed r , [lemma 3.1](#) gives $K_r/K_2^{r/2} = O_r(m^{-(r-2)/2})$. The central interval $|\theta| \leq t \log m/\sqrt{m}$ corresponds to $|v| \leq C \log m$. Taylor estimates for $H(t - i\theta)$ follow directly from its absolutely convergent sums: the absolute value of a derivative along this vertical segment is bounded by the corresponding positive derivative sum at t .

Expand the exponential in (3.16), assigning weight $r - 2$ to the term of degree r . Retain total weight through $2J + 1$. The integrated terms of odd weight vanish because their total degree in v is odd. The weight- $2j$ terms have Gaussian integral $(-1)^{R/2}(R - 1)!!$ after division by $\sqrt{2\pi}$, giving (3.8).

For completeness, the remainder estimate does not require retaining a power of $\log m$. On $|v| \leq C \log m$, the absolute correction to the Gaussian exponent is $O(m^{-1/2}(|v|^3 + 1))$, which is at most $v^2/4 + o(1)$ outside a fixed bounded interval. A Taylor remainder through weight $2J + 1$ is bounded by

$$C_J m^{-J-1} (1 + |v|^{M_J}) e^{-v^2/4}$$

for a fixed integer M_J , after including the Gaussian factor and increasing the Taylor order of H to $2J + 4$ if necessary. Its integral is $O_J(m^{-J-1})$. The omitted Gaussian tails are superpolynomially small. Combining this with (3.13)–(3.15) proves [theorem 3.2](#).

The leading values of the correction polynomials are useful checks:

$$E_1 = -\frac{1}{12m} + O(m^{-2}), \quad E_2 = \frac{1}{288m^2} + O(m^{-3}), \quad E_3 = \frac{139}{51840m^3} + O(m^{-4}). \quad (3.17)$$

They follow by substituting $B_r \sim (r - 1)!m$ in (3.8). At a finite m , using the exact cumulants is preferable to replacing each E_j by only its first universal term.

3.4 A reusable sufficient hypothesis

The same proof applies to an increasing sequence of positive integers $a_1, a_2, \dots$ when $a_1 = 1$, the ratios are eventually between two fixed constants $1 < q \leq Q < \infty$, and a finite subset of $\{a_2, a_3, \dots\}$ has greatest common divisor one. Replace m by the number of active parts $a_k \leq 1/t$, up to an $O(1)$ difference. The finite gcd condition is sufficient for the last compact arc estimate; it is not claimed to be necessary. In particular, the proof as stated should not be applied unchanged to pure powers of two, whose tail has a nontrivial gcd.

4 The Catalan product and its moving periodic phase

4.1 The logarithm of a Catalan number and the Barnes product

The standard gamma quotient in (3.3) gives the complete expansion

$$r(x) = \alpha x - \beta \log x + c + \sum_{j \geq 1} b_j x^{-j}, \quad \beta = \frac{3}{2}, \quad c = -\frac{1}{2} \log \pi, \quad b_1 = -\frac{9}{8}, \quad b_2 = \frac{1}{2}. \quad (4.1)$$

The meaning is asymptotic, to every fixed order with differentiated remainders. A convenient exact coefficient formula is

$$b_j = \frac{(-1)^{j+1}}{j(j+1)} (B_{j+1}(1/2) - B_{j+1}(2)), \quad (4.2)$$

where $B_j(x)$ is a Bernoulli polynomial. For example, $b_3 = -21/64$. This follows by subtracting the shifted Stirling expansions of the two log-gamma functions; it also gives an effective coefficient source for every subsequent step.

For an integer $M \geq 0$, put $S(M) = \sum_{k=1}^M r(k)$. Repeated use of $G(z+1) = \Gamma(z)G(z)$ gives the exact identity

$$S(M) = \frac{\alpha}{2} M(M+1) - \frac{M}{2} \log \pi + \log G(M+3/2) - \log G(M+3) - \log G(3/2). \quad (4.3)$$

The Barnes expansion [5], or Euler–Maclaurin summation followed by the exact identity to fix the constant, yields

$$S(M) = \frac{\alpha}{2} M^2 - \frac{3}{2} M \log M + \left(\frac{\alpha}{2} - \frac{1}{2} \log \pi + \frac{3}{2} \right) M - \frac{15}{8} \log M + K \\ - \frac{19}{16M} + \frac{65}{128M^2} + O(M^{-3}), \quad (4.4)$$

$$K = -\frac{3}{4} \log(2\pi) - \log G(3/2). \quad (4.5)$$

Subtracting the expansion at $M-1$ from the expansion at M recovers (4.1); this is a useful local check, but the Barnes identity is what determines the constant K .

4.2 Separating the bulk from a localized lattice sum

Let

$$g(v) = -\log(1 - e^{-v}) - (-v)_+, \quad (-v)_+ = \max\{-v, 0\}. \quad (4.6)$$

As $v \rightarrow -\infty$, $g(v) = O(e^v)$; as $v \rightarrow +\infty$ it decays superexponentially. It is continuous and piecewise analytic, with a jump in its first derivative at zero. This mild kink will cancel against the fractional-part polynomial.

For $m \geq 1$, let $M = \lfloor m \rfloor$ and $\theta = m - M$. Define

$$\mathcal{H}(m) = H(e^{-r(m)}). \quad (4.7)$$

Then the following identity is exact:

$$\mathcal{H}(m) = Mr(m) - S(M) + \sum_{k \geq 1} g(r(k) - r(m)). \quad (4.8)$$

Indeed, the subtracted linear part $(-v)_+$ contributes $r(m) - r(k)$ precisely for $k \leq M$. If m is an integer, the boundary term is zero, so no convention ambiguity occurs.

Replacing $r(m + d) - r(m)$ by αd in the localized sum suggests

$$\Psi_\alpha(\theta) = \frac{\alpha}{2}\theta(1 - \theta) + \sum_{j \in \mathbb{Z}} g(\alpha(j - \theta)), \quad 0 \leq \theta \leq 1. \quad (4.9)$$

The sum converges absolutely, exponentially fast on the left and faster on the right. The polynomial is forced by expansion of $Mr(m) - S(M)$; it is not an adjustable normalization.

Lemma 4.1 (Fourier representation of the phase). *The lattice formula (4.9) equals (2.3). In particular it extends to a smooth, nonconstant one-periodic function, and termwise derivatives of its Fourier series of every fixed order are legitimate.*

Proof. For $\operatorname{Re} s > 0$, change variables $y = e^v$ and expand $-\log(1 - e^{-y}) = \sum_{\ell \geq 1} e^{-\ell y} / \ell$ to obtain

$$\int_{\mathbb{R}} g(v) e^{sv} dv = \Gamma(s) \zeta(1 + s) - \frac{1}{s^2}.$$

The integral involving g extends analytically to $\operatorname{Re} s > -1$. The apparent singularity on the right at zero is removable: the product of the Laurent expansions of $\Gamma(s)$ and $\zeta(1 + s)$ has leading term s^{-2} , no s^{-1} term, and constant term κ . Consequently the mean of the lattice sum is κ/α .

For $\ell \neq 0$, integrating its Fourier coefficient over $0 \leq \theta \leq 1$ and unfolding the lattice gives

$$\frac{1}{\alpha} \left\{ \Gamma(s_\ell) \zeta(1 + s_\ell) - \frac{1}{s_\ell^2} \right\}, \quad s_\ell = \frac{2\pi i \ell}{\alpha}.$$

The coefficient of $\alpha\theta(1 - \theta)/2$ is $-\alpha/(4\pi^2\ell^2)$, which cancels the contribution of $-1/(\alpha s_\ell^2)$. Its mean is $\alpha/12$. This proves (2.3). The gamma factor has exponential decay on the vertical axis; the zeta factor and its fixed-order derivatives have at most polynomial growth there, as follows for example from Euler–Maclaurin continuation. Hence every differentiated Fourier series converges absolutely. Nonconstancy was proved in [corollary 2.2](#). $\square$

[Write note, batch 7301, 1 October 2026.] The same Mellin kernel and the same constant occur elsewhere in ProveIt, for a different product. The Fabius-tree compilation *Geometric Uniform Convolutions and New Frontiers* [10], in its section “Mellin analysis and the hidden log-periodic phase”, treats the Laplace product $H_q(z) = \prod_{k \geq 0} (1 - e^{-zq^k}) / (zq^k)$, with kernel $g(x) = \log((1 - e^{-x})/x)$, Mellin transform $-\Gamma(1 + s)\zeta(1 + s)/s = -\Gamma(s)\zeta(1 + s)$, and constant $c_1 = \pi^2/12 - \gamma^2/2 - \gamma_1$, which is the κ of (2.2). Neither text proves the other’s theorem: the products, the lattices and the normalizations differ, and the coincidence is the de Bruijn–Mahler mechanism common to both.

4.3 The first two bulk orders, with a uniform remainder

Theorem 4.2 (Moving-lattice expansion). *Uniformly in the fractional part of m , as $m \rightarrow \infty$,*

$$\mathcal{H}(m) = \frac{\alpha}{2}m^2 - \left(\frac{3}{2} + \frac{\alpha}{2}\right)m + \frac{15}{8}\log m - \frac{9}{8} - K + \Psi_\alpha(m) + \frac{P_1(m)}{m} + O(m^{-2}), \quad (4.10)$$

$$P_1(\theta) = \frac{27}{16} - \frac{3}{2}\partial_\alpha \Psi_\alpha(\theta). \quad (4.11)$$

There is a complete extension with $\sum_{j=1}^J P_j(m)m^{-j}$ and remainder $O_J(m^{-J-1})$. The P_j are smooth and one-periodic. The expansion may be differentiated any fixed number of times if sufficiently many terms are retained.

Proof. In (4.8), put $d = k - m$. For a bounded d ,

$$r(m+d) - r(m) = \alpha d - \frac{\beta d}{m} + O((1+|d|)^2 m^{-2}).$$

Taylor expansion of the localized term therefore begins as

$$g(\alpha d) - \frac{\beta d}{m}g'(\alpha d) + O(m^{-2}(1+|d|)^L e^{-c_0|d|})$$

for a fixed L ; on the positive side an even stronger tail is available. One-sided values at $d = 0$ cause no difficulty because the multiplier d vanishes. The missing terms with $k \leq 0$ in the infinite lattice are exponentially small.

Expansion of the exact bulk term using (4.1) and (4.4) gives

$$\begin{aligned} Mr(m) - S(M) &= \frac{\alpha}{2}m^2 - (\beta + \alpha/2)m + (\beta/2 - b_1)\log m + b_1 - K \\ &\quad + \frac{\alpha}{2}\theta(1-\theta) + \frac{27/16 - (\beta/2)\theta(1-\theta)}{m} + O(m^{-2}). \end{aligned}$$

Combining this with the localized sum gives (4.10), since

$$\partial_\alpha \Psi_\alpha(\theta) = \frac{1}{2}\theta(1-\theta) + \sum_{j \in \mathbb{Z}} (j-\theta)g'(\alpha(j-\theta)).$$

Here is a uniform justification of the arbitrary-order procedure. Split the lattice into $|d| \leq m^{1/3}$ and its complement. In the first range, expansions of $r(m+d) - r(m)$ and Taylor expansions of g have remainders bounded by a fixed polynomial in $|d|$ times $m^{-J-1}e^{-c_0|d|}$. The signs of d and of the two arguments of g coincide, so Taylor expansion never crosses the kink. Summing preserves $O_J(m^{-J-1})$. In the complementary range, monotonicity and the positive lower bound for r' give an exponentially small tail, including the initial indices far to the left. The same reasoning applies after any fixed number of derivatives, using more terms at the start. At integral m , the jumps of the one-sided lattice derivatives cancel exactly with those of the fractional-part bulk polynomials. This can be seen either from the exact decomposition, by transferring the boundary index between the two pieces, or coefficient by coefficient in that identity. The resulting coefficient functions therefore match with all fixed-order derivatives across the endpoints of a period. $\square$

4.4 A constructive rule for every coefficient

The preceding proof can be made into a finite symbolic recipe. Write

$$r(m+d) - r(m) = \alpha d + \sum_{j \geq 1} D_j(d)m^{-j}, \quad (4.12)$$

where

$$D_j(d) = \frac{\beta(-1)^j d^j}{j} + \sum_{h=1}^{j-1} b_h (-1)^{j-h} \binom{j-1}{h-1} d^{j-h}. \quad (4.13)$$

For example, $D_1(d) = -\beta d$ and $D_2(d) = \beta d^2/2 - b_1 d$. The localized contribution to the coefficient of m^{-j} is

$$T_j(\theta) = \sum_{d \in \mathbb{Z}-\theta} \sum_{\ell=1}^j \frac{g^{(\ell)}(\alpha d)}{\ell!} \sum_{\substack{r_1+\dots+r_\ell=j \\ r_\nu \geq 1}} \prod_{\nu=1}^{\ell} D_{r_\nu}(d). \quad (4.14)$$

Each coefficient is thus an absolutely convergent lattice sum of explicit functions. At $d = 0$, all $D_j(0)$ vanish, so arbitrary consistent one-sided values of the derivatives give the same term, namely zero.

Expand $Mr(m) - S(M)$ with $M = m - \theta$, then add (4.14). This produces $P_j(\theta)$. Bernoulli polynomials supply the bulk coefficients; finite compositions of the D_j supply the localized coefficients. This is an all-orders construction, not merely a claim that unspecified coefficients exist. Fourier transformation can be used afterward to replace the localized sums by rapidly convergent Fourier series.

5 Deriving the explicit coefficient expansion

5.1 The saddle coordinate and its correction

Let $t = e^{-r(m)}$, and define

$$Q(m) = t K_1(t) = \frac{\mathcal{H}'(m)}{r'(m)}. \quad (5.1)$$

Differentiation of (4.10), followed by series division, gives

$$Q(m) = m + q_0(m) + \frac{q_1(m)}{m} + O(m^{-2}), \quad (5.2)$$

$$q_0 = -\frac{1}{2} + \frac{\Psi'_\alpha}{\alpha}, \quad q_1 = \frac{P'_1}{\alpha} + \frac{\beta \Psi'_\alpha}{\alpha^2}. \quad (5.3)$$

The phase functions on the right are evaluated at m . The scaled variance satisfies the exact identity

$$B_2 = t^2 K_2 = Q + \frac{Q'}{r'}, \quad (5.4)$$

so that

$$B_2 = m + q_0 + \frac{1 + q'_0}{\alpha} + O(m^{-1}). \quad (5.5)$$

More generally, the recurrence

$$B_{j+1} = j B_j + \frac{B'_j}{r'} \quad (5.6)$$

generates every scaled cumulant from Q . This is useful both for the proof and for the all-orders construction.

At the saddle, $n = e^{r(m)} Q(m)$. Consequently

$$\log n = \alpha m - \frac{1}{2} \log m + c + \frac{b_1 + q_0(m)}{m} + \frac{b_2 + q_1(m) - q_0(m)^2/2}{m^2} + O(m^{-3}). \quad (5.7)$$

The core coordinate (2.5) removes precisely the first three terms. Write

$$m = \mu + \frac{a(\mu)}{\mu} + \frac{b(\mu)}{\mu^2} + O(\mu^{-3}). \quad (5.8)$$

Taylor substitution into (5.7) gives the explicit functions

$$a = -\frac{b_1 + q_0}{\alpha}, \quad (5.9)$$

$$b = \frac{(1/2 - q'_0)a - (b_2 + q_1 - q_0^2/2)}{\alpha}. \quad (5.10)$$

All functions here are evaluated at μ . The residual derivative in (5.7) is $\alpha + O(1/m)$, which is bounded away from zero. Thus the formal coefficient comparison also gives the asserted analytic error by the mean value theorem. Repeating the argument proves an all-orders reversion with smooth periodic coefficients.

5.2 The normalization and the first correction

Taking logarithms of (3.7), using (3.17), and writing $K_2 = e^{2r(m)} B_2$ gives

$$\log p(n) = \mathcal{H}(m) + Q(m) - r(m) - \frac{1}{2} \log(2\pi B_2(m)) - \frac{1}{12m} + O(m^{-2}). \quad (5.11)$$

Let $d = b_1 + q_0$. Substituting the preceding expansions into this equation first produces the expression in the *true* saddle coordinate:

$$\log p(n) = \frac{\alpha}{2} m^2 - Am + \eta \log m + C + \Psi_\alpha(m) + d(m) + \frac{U(m)}{m} + O(m^{-2}), \quad (5.12)$$

$$U = P_1 + q_1 - b_1 - \frac{q_0}{2} - \frac{1 + q'_0}{2\alpha} - \frac{1}{12}. \quad (5.13)$$

The constant simplifies because

$$-K - c - \frac{1}{2} \log(2\pi) = \frac{1}{4} \log 2 + \frac{3}{4} \log \pi + \log G(3/2) = C.$$

Now substitute (5.8). At order one, $\alpha a + d = 0$, so the apparently additional phase derivative in d *cancels*. At order $1/\mu$, the coefficient is

$$\alpha b + (-A + \Psi'_\alpha + q'_0)a + U. \quad (5.14)$$

Using (5.9)–(5.10), this simplifies first to

$$P_1 - \frac{7}{12} - \frac{1}{2\alpha} + (1/2 - b_1)q_0 - \frac{1}{2}q_0^2 - \frac{q'_0}{2\alpha},$$

and then, with $b_1 = -9/8$ and (4.11), exactly to (2.4). In particular the P'_1 terms disappear. This explicit cancellation is one reason to distinguish m and μ throughout. Equations (2.6) and (2.7) follow.

5.3 Completion of the all-orders proof

For a fixed requested order J , use the following finite operations. First compute sufficiently many b_j from (4.2), bulk coefficients from (4.3), and P_j from (4.14). Next form Q and $B_2, B_3, \dots$ by (5.1) and (5.6). Substitute these in the finite Gaussian polynomials (3.8) and take the logarithm of the saddle prefactor and correction. Finally revert (5.7) about μ and substitute.

The derivative-controlled estimates in theorem 4.2 justify each finite calculus operation. The saddle theorem supplies the coefficient-extraction remainder. The reversion has derivative bounded away from zero, so it preserves the desired error once one extra order has been retained before substitution into the quadratic term. All coefficient operations are addition, multiplication, differentiation, and division by series with nonzero constant term in an algebra of smooth periodic functions. Thus no new unbounded coefficient, such as $\log \mu$, is generated beyond the single displayed $\eta \log \mu$. This proves (2.8) and completes theorem 2.1.

A safe implementation rule is to retain two orders more than the final requested order during the m -to- μ substitution. This is a bookkeeping rule, not a new analytic assumption.

6 Inverse growth and the integer staircase

6.1 Reversion of the quadratic logarithmic core

Start with $Y = \log y$ and $s = \sqrt{2Y/\alpha}$. The leading quadratic term suggests $\mu = s + A/\alpha + o(1)$. Set

$$\mu = s + \frac{A}{\alpha} + \frac{v(s)}{s} + O\left(\frac{(\log s)^2}{s^2}\right).$$

Substitution into (2.6), at constant order after the quadratic and linear terms cancel, yields

$$v(s) = \frac{A^2}{2\alpha^2} - \frac{\eta \log s + C + \Psi_\alpha(s + A/\alpha)}{\alpha}. \quad (6.1)$$

Since $\log n = \alpha\mu - \frac{1}{2} \log \mu - \frac{1}{2} \log \pi$, (2.12) follows by Taylor expansion. The term $D_\alpha(\mu)/\mu$ in $\log p$ first affects the next inverse order, not the displayed $1/R$ correction.

Higher inverse terms are obtained by the same procedure. The leading inverse derivative is of size s , so a residual term is removed by division by s and a bounded perturbation. Every differentiation of a periodic coefficient leaves it in the same coefficient algebra, while products of the already occurring $\log s$ terms produce polynomials in $\log s$. The phase is always reducible by Taylor expansion to $s + A/\alpha$. This proves the asserted form of the all-orders inverse.

6.2 Why a continuous calculation describes the discrete threshold

Adding a unit part proves that $p(n)$ is nondecreasing. Moreover, $p(n) - p(n-1)$ counts partitions of n without a unit part. Every integer $n \geq 4$ is a nonnegative combination of 2 and 5: even n use only 2, and odd $n \geq 5$ use one 5 and then 2. Therefore $p(n) > p(n-1)$ for all $n \geq 4$, and the threshold (2.10) is unambiguous for large y .

Let $f_J(x)$ denote the smooth right side of (2.8), with n replaced by a large positive real x . Its derivative with respect to $X = \log x$ satisfies

$$\frac{d}{dX} f_J(e^X) = \mu(e^X) + O(1) > 0. \quad (6.2)$$

The asymptotic theorem says, at integers, $\log p(n) = f_J(n) + O(\mu(n)^{-J-1})$ with a uniform constant. If $f_J(x) = Y$, a change of $O(\mu^{-J-2})$ in $\log x$ moves $f_J(x)$ by the entire coefficient error. Bracketing the corresponding real arguments by neighboring integers therefore gives

$$\log N(y) - \log x = O(\mu(x)^{-J-2}) + O(x^{-1}), \quad (6.3)$$

provided x is the inverse of the chosen smooth truncation. The bracket uses only monotonicity, not an assumed canonical interpolation of p .

The $O(x^{-1})$ term comes from rounding an argument to an integer; it is smaller than every fixed inverse power of R . The first term is the analytic truncation uncertainty and is much larger at a fixed order. Accordingly, the theorem gives an asymptotic formula for the threshold, not an exact integer reconstruction algorithm for every y . Equations (6.1) and (6.3) prove [theorem 2.4](#).

6.3 A practical caution

The expansion parameter is approximately $1/\log n$, not $1/n$. Even at $n = 10^6$, the effective number of active parts is only about 11.36. The leading inverse equivalent can therefore have a large finite-argument error. The first correction helps substantially, but a numerical application at moderate size should use the implicit saddle formula and invert it numerically, preferably with a certified remainder if an integer decision is required. [Section 9](#) reports actual errors rather than presenting an asymptotic equivalent as a uniformly accurate approximation.

7 The largest Catalan part in a random partition

Choose uniformly among the $p(n)$ partitions of n . Let J_n be the largest index k for which a part of size c_k occurs. For $n > 0$ this is well-defined. It is natural to center J_n at the saddle cutoff, not at $\log_4 n$ alone.

Theorem 7.1 (A phase-dependent largest-index law). *Let $m = m_n$ be the true saddle coordinate, $M = \lfloor m \rfloor$, and $\theta_n = m - M$. For every fixed integer h ,*

$$\mathbb{P}(J_n \leq M + h) = \prod_{j=h+1}^{\infty} (1 - \exp\{-4^{j-\theta_n}\}) + O_h(m_n^{-1}). \quad (7.1)$$

In particular, along any subsequence with $\theta_n \rightarrow \theta \in [0, 1)$, $J_n - M$ converges in distribution to the integer-valued law with cumulative probabilities

$$G_h(\theta) = \prod_{j=h+1}^{\infty} (1 - e^{-4^{j-\theta}}). \quad (7.2)$$

The estimates are for fixed h ; a growing- h or moment theorem is not being asserted.

Proof. Under the product probability measure at parameter t , let the numbers of parts of sizes c_k be independent geometric random variables:

$$\mathbb{P}(Z_k = z) = (1 - e^{-tc_k})e^{-tc_k z}, \quad z = 0, 1, 2, \dots$$

Conditioning on $\sum_k c_k Z_k = n$ gives the uniform distribution on partitions of n , since every such partition has the same product weight. This provides an exact probabilistic interpretation, not an approximation.

Put $K = M + h$ and split the product at K . Let H_{pre} and H_{tail} be its logarithms. At the full saddle, the removed tail has scaled cumulants of every fixed order $O_h(1)$. To see this, write $k = M + j$ and use

$$tc_{M+j} = \exp\{r(M+j) - r(m)\},$$

which is bounded away from zero for each fixed starting $j = h + 1$ and then grows geometrically. The corresponding sums of the functions f_r from [lemma 3.1](#) are bounded. Thus the removed mean is $O_h(1/t)$, whereas the full standard deviation is $\asymp \sqrt{m}/t$.

Apply the same coefficient saddle argument to the prefix product. Its saddle differs from t by $O_h(t/m)$: the mean deficit is $O_h(1/t)$ and the derivative of the mean is of size m/t^2 . Taylor's theorem then shows that its saddle action differs from the prefix action evaluated at t by $O_h(1/m)$. The variance ratio is $1 + O_h(1/m)$, and the two first saddle correction factors are also $1 + O_h(1/m)$. The arc estimates remain uniform because the prefix still contains $m - O_h(1)$ active parts and all the fixed small factors used in [Section 3](#). Consequently,

$$\frac{[z^n] \prod_{k \leq K} (1 - z^{c_k})^{-1}}{p(n)} = e^{-H_{\text{tail}}(t)} (1 + O_h(m^{-1})) = \prod_{k > K} (1 - e^{-tc_k}) (1 + O_h(m^{-1})).$$

This ratio is exactly the desired conditional probability.

Finally, for each fixed j ,

$$tc_{M+j} = 4^{j-\theta_n} \left(1 + O\left(\frac{1+|j|}{m}\right) \right).$$

For $0 \leq j - \theta_n \leq m/2$, the corresponding logarithmic error is bounded by $C(1 + |j|)/m$. The derivatives of the logarithmic tail factors have a summable, superexponentially decaying majorant throughout this range. The range $j - \theta_n > m/2$ contributes a superexponentially small tail, and the finitely many negative indices are controlled separately. Thus summation replaces the tail product by (7.2) with error $O_h(m^{-1})$. This proves (7.1). For each phase, $G_h \rightarrow 1$ as $h \rightarrow +\infty$ and $G_h \rightarrow 0$ as $h \rightarrow -\infty$, so it is the cumulative distribution of a proper integer-valued law. Pointwise convergence at the integer cutoffs gives the distributional assertion. $\square$

The local scale $4^{j-\theta}$ in this law is the same scale that produces the Fourier amplitude in (2.3). Both results describe the bounded-width transition between parts that are typically available in many copies and parts that are rarely used. The phase is not noise: it records where the continuous cutoff falls between two consecutive Catalan indices.

8 A universality statement for polynomially perturbed geometric parts

The preceding method separates universal local behavior from sequence-specific normalization. The following proposition makes that separation precise.

Proposition 8.1 (General exponential–polynomial part growth). *Suppose an increasing integer part sequence a_k satisfies the sufficient saddle hypotheses at the end of Section 3. Suppose its logarithm has a smooth eventual interpolation with a termwise differentiable expansion*

$$r(k) = \alpha k - \beta \log k + c + b_1/k + b_2/k^2 + \cdots, \quad \alpha > 0.$$

Define $K_{\mathcal{A}}$ by

$$\begin{aligned} \sum_{k=1}^M \log a_k &= \frac{\alpha}{2} M^2 - \beta M \log M + (\alpha/2 + c + \beta) M \\ &\quad + (b_1 - \beta/2) \log M + K_{\mathcal{A}} + O(M^{-1}). \end{aligned}$$

Let $p_{\mathcal{A}}(n)$ be the corresponding partition number, and let μ be the large solution of $n = e^c e^{\alpha\mu} \mu^{1-\beta}$. Then

$$\begin{aligned} \log p_{\mathcal{A}}(n) &= \frac{\alpha}{2} \mu^2 + (1 - \beta - 3\alpha/2) \mu + (3\beta/2 - b_1 - 1/2) \log \mu \\ &\quad - K_{\mathcal{A}} - c - \frac{1}{2} \log(2\pi) + \Psi_{\alpha}(\mu) + O(\mu^{-1}). \end{aligned} \tag{8.1}$$

There is an all-orders extension with smooth periodic inverse-power coefficients. The function Ψ_{α} is exactly (2.3); it depends on α , not on β or the finite initial segment.

Proof. The constant $K_{\mathcal{A}}$ exists by Euler–Maclaurin summation of the assumed expansion, or by subtracting enough explicitly summable terms. The bulk and local argument gives

$$\mathcal{H}(m) = \frac{\alpha}{2} m^2 - (\beta + \alpha/2)m + (\beta/2 - b_1) \log m + b_1 - K_{\mathcal{A}} + \Psi_{\alpha}(m) + O(m^{-1}).$$

The local limit $r(m+d) - r(m) \rightarrow \alpha d$ is unchanged. Also $Q = m + q_0 + O(m^{-1})$, where $q_0 = -1/2 + \Psi'_{\alpha}/\alpha$, independent of β . Therefore

$$\log n = \alpha m + (1 - \beta) \log m + c + (b_1 + q_0)/m + O(m^{-2}).$$

Reversion about μ again gives $m = \mu - (b_1 + q_0)/(\alpha\mu) + O(\mu^{-2})$. In the logarithm of the saddle estimate, the constant $b_1 + q_0$ cancels against the quadratic-core displacement just as before. The remaining linear, logarithmic, and constant terms are exactly (8.1). The proof of every higher order is the same finite construction with the assumed coefficients b_j in place of their Catalan values. $\square$

This proposition explains why merely computing the exponential ratio of part sizes is insufficient. The coefficient β changes the linear term and the size coordinate; b_1 changes the power of μ ; and the whole initial sequence contributes to $K_{\mathcal{A}}$. The local periodic shape, on the other hand, is determined by the limiting logarithmic gap.

9 Computation, independent checks, and reproducibility

9.1 What the supplied programs do

The package contains an exact integer dynamic program for $p(0), \dots, p(N)$, a separate logarithmic-derivative recurrence check, arbitrary-precision saddle evaluation, independent Fourier and real-lattice evaluations of the phase, the explicit first-correction formula, and the inverse formula. The numerical program implements E_1 and E_2 directly; a rational multi-index generator lists the Gaussian polynomials through E_4 in the recorded data. The all-orders theorem does not depend on a claim that an unlimited symbolic implementation is already supplied.

Exact dynamic programming starts with $p(0) = 1$ and $p(n) = 0$ for $n > 0$. For each allowed part $c \leq N$, in increasing order, replace $p(n)$ by $p(n) + p(n - c)$ for $n = c, c + 1, \dots, N$. The ascending update allows repeated copies of c . An independent recurrence is

$$np(n) = \sum_{j=1}^n b(j)p(n-j), \quad b(j) = \sum_{c_k | j} c_k, \quad (9.1)$$

obtained from $zF'(z)/F(z)$. Agreement of these two algorithms is checked through $n = 300$. The first 20 terms agree with the inspected OEIS entry. The stored small coefficient file is independently generated; it is not a redistributed download of another author's table.

The complete run used Python 3.13.5 and mpmath 1.3.0 at 60 decimal digits. Exact coefficients were generated through $N = 10^6$. The Barnes product identity was checked at $M = 0, \dots, 20$. Fourier and real-lattice formulas for both Ψ_α and its partial α -derivative were compared at 10 phases; their maximum recorded discrepancy was less than $4.8 \cdot 10^{-59}$. Rational Gaussian algebra gives the first three constants in (3.17) exactly.

These statements describe what was actually executed. Decimal output and finite tail cutoffs are not rigorous interval enclosures, and the tests do not prove the asymptotic estimates. A separate symbolic audit verifies the normalization and first-correction cancellation using exact algebra.

9.2 Coefficient accuracy

Let P_0 denote the saddle prefactor in (3.7), $P_1 = P_0(1 + E_1)$, and $P_2 = P_0(1 + E_1 + E_2)$ in this subsection only. The table reports signed relative error, $P_j/p(n) - 1$.

Table 1: Exact-cumulant saddle approximations. Errors are not assumed to be monotone.

n	m_n	P_0 error	P_1 error	P_2 error
10^2	4.5337	$2.126 \cdot 10^{-2}$	$-4.169 \cdot 10^{-3}$	$-3.016 \cdot 10^{-3}$
10^3	6.2400	$1.832 \cdot 10^{-2}$	$7.139 \cdot 10^{-4}$	$4.696 \cdot 10^{-4}$
10^4	7.9492	$1.380 \cdot 10^{-2}$	$2.681 \cdot 10^{-4}$	$3.448 \cdot 10^{-5}$
10^5	9.6549	$9.577 \cdot 10^{-3}$	$-3.205 \cdot 10^{-4}$	$-1.160 \cdot 10^{-4}$
10^6	11.3564	$8.557 \cdot 10^{-3}$	$6.471 \cdot 10^{-5}$	$4.661 \cdot 10^{-5}$

For reference, some exact values from the computation are

$$\begin{aligned} p(100) &= 2031, \\ p(1000) &= 18299196, \\ p(10000) &= 8125868781802, \\ p(100000) &= 180507423003606259924, \\ p(1000000) &= 198409222879647727339514603123. \end{aligned}$$

The explicit μ formula is more transparent but converges more slowly at these arguments. Exponentiating the right side of (2.6) without, and with, D_α/μ gives the following signed relative errors.

Table 2: Explicit periodic formulas at moderate arguments.

n	Through the periodic constant term	Including D_α/μ
10^2	-0.122792	-0.069993
10^3	-0.072947	-0.036697
10^4	-0.048618	-0.018121
10^5	-0.040289	-0.012929
10^6	-0.032146	-0.010559

As a check independent of coefficient extraction, substitute $m = 10.2, 30.2, 100.2, 300.2$ directly into the exact product logarithm and subtract (4.10) through P_1/m . The residual multiplied by m^2 is respectively about $-0.1474, -0.2747, -0.3078, -0.3166$. This is consistent with the next periodic coefficient at the fixed phase 0.2. It is not used to infer the remainder theorem.

9.3 Inverse and largest-index checks

At $y = p(n)$ and $n \geq 4$, the true threshold is exactly $N(y) = n$. This permits an unambiguous inverse test.

Table 3: Signed relative error of the inverse approximations at $y = p(n)$.

n	Leading equivalent (2.13)	Corrected formula (2.12)
10^3	2.042368	0.133978
10^4	1.548609	0.071730
10^5	1.256439	0.044236
10^6	1.061185	0.029094

The large leading errors are displayed deliberately: an asymptotic equivalent can be mathematically correct while still being a poor numerical approximation at a million. The corrected formula is much better in this test, but is not a certified integer inversion rule.

At $n = 10^6$ we have $M = 11$ and $\theta_n \approx 0.35639211$. Prefix dynamic programs give exact numerators for the event in theorem 7.1. The comparison is The exact probabilities in the table mean ratios of exactly

Table 4: Largest-index probabilities at $n = 10^6$.

h	Cutoff $M + h$	Exact probability	$G_h(\theta_n)$
-2	9	0.05550439	0.05898171
-1	10	0.39681499	0.41691590
0	11	0.89851234	0.91283590
1	12	0.99999711	0.99994242

computed integers, displayed as decimals. Their limits retain the phase because only a bounded number of indices occupy the transition region.

9.4 Numerical safeguards and build instructions

The cumulant implementation works with the scaled quantities $B_j = t^j K_j$ to avoid unnecessary overflow. It evaluates $1 - e^{-y}$ using `-expm1(-y)` near zero. The geometric sums in (3.1) are computed as rational functions using Eulerian polynomials, rather than by summing a slowly convergent inner series.

For the Fourier coefficients, the supplied code explicitly requests the Euler–Maclaurin method for $\zeta(1 + 2\pi i \ell / \log 4)$ and its derivative. At even ℓ , the denominator $1 - 2^{1-s}$ of an eta-based formula for $\zeta(s)$ vanishes at the evaluation point, so a naive numerical quotient is ill-conditioned. In the tested mpmath environment, forcing the Euler–Maclaurin route removed a discrepancy exposed by the independent real-lattice check. This is a safeguard for the tested implementation, not a claim about every current zeta library. The relevant evaluation methods are documented by mpmath [8].

From the package directory, run

```
python code/verify.py --max-n 10000000 --dps 60
python code/symbolic_audit.py
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The CSV files are convenient for inspecting the tables, and `data/verification.json` records the environment, exact checks, constants, and additional Gaussian coefficients. The code performs no network access. The bibliography and source-audit notes record the external provenance separately.

[Write note, batch 7301, 1 October 2026.] In the collection these commands must not be run inside the report directory. `code/verify.py` writes by default to the `data/` directory beside `code/`, that is to the shipped data, unless `-output` names another directory; `code/symbolic_audit.py` always writes `../data/symbolic_audit.json` relative to its own location and has no option; and `code/build.sh` changes into `code/`, where there is no `article.tex`, so it fails as shipped. Rerun on a copy of the report directory (the collection README gives the commands). At intake both scripts were rerun on such a copy: the three CSV files were byte-identical to the delivery, `a033552_0_300.txt` and `symbolic_audit.json` equal apart from Windows line ends, and `verification.json` equal apart from line ends and the recorded Python and platform strings.

10 Further research questions and concrete next steps

The following directions are deliberately separated from the proved statements. Each would improve either the mathematical scope or the reliability of computation.

10.1 Certified errors and exact inverse decisions

Replace the unspecified constants in the contour and moving-lattice remainders by explicit, computable bounds. The proof already divides the problem into positive sums, elementary arc inequalities, and Taylor remainders. An interval implementation could enclose the saddle, truncate both lattice tails with proven bounds, and return a rigorous interval for $\log p(n)$. The decisive inverse question is then whether that interval separates two neighboring integer thresholds. This is substantially stronger than a fixed-order asymptotic equivalent.

10.2 A symbolic compiler for periodic asymptotic coefficients

Implement (4.14), (5.6), (3.8), and the m -to- μ reversion in one coefficient algebra. A useful output would include L_2, L_3, L_4 in a normalized basis of phase derivatives, together with exact rational audits. The central design issue is to avoid treating $\Psi_\alpha(m)$ as a constant under differentiation and to track the partial α -derivative separately from the phase derivative.

10.3 Beyond-all-orders contributions

The proof discards the compact outer arcs at a relative scale $O(t\sqrt{m})$, much smaller than every inverse power of m . Their actual contributions may have an arithmetic organization at roots of unity. Identify the dominant secondary arcs, their exponential actions, and how they interact with the analytic tail of the Barnes expansion. A complete transseries including such sectors would go beyond the all-orders Poincaré expansion established here.

10.4 Joint statistics near the largest part

For finitely many indices $M + j$, the unconditioned geometric variables suggest a phase-dependent local product limit. Prove a joint conditional limit with an explicit first correction, rather than only the tail-absence probability in [theorem 7.1](#). Extend the estimates uniformly in h far enough to justify moment convergence and asymptotics for the expected largest index. Neither extension follows automatically from the fixed- h statement.

10.5 The number of parts and mixed conditioning

Introduce a marking variable for the total number of parts. Small allowed parts and the bounded-width cutoff have very different fluctuation scales. Determine the correct normalization and a joint limit with J_n . A two-variable saddle may need a different treatment from a direct Gaussian approximation because the unit-part multiplicity can remain macroscopic on the scale $1/t$.

10.6 Fuss–Catalan and other gamma-quotient part sequences

Apply [proposition 8.1](#) to integer part sequences with a gamma-quotient asymptotic, after checking the initial-part gcd condition and removing any duplicated values. Calculate the normalization constant by Barnes or multiple-gamma products when possible. The proposition predicts a universal periodic shape for each limiting logarithmic gap, but the power and the normalization require a new calculation in every family.

10.7 Exact counting complexity

The supplied dynamic program uses $O(n \log n)$ integer additions to compute all coefficients through n , because there are $O(\log n)$ allowed parts up to n . That is not a polynomial-in- $\log n$ algorithm for a single exact value. Determine whether Catalan-part partitions admit a substantially faster exact counting method of the kind motivated by Pak’s discussion [2]. The asymptotic formulas may provide useful bounds or numerical guidance, but they do not settle this question.

10.8 Formalization and a periodic coefficient algebra

The finite coefficient identities, the rational Gaussian multi-index formula, the gamma-quotient recurrence for Catalan numbers, and the monotonicity argument for $N(y)$ are natural first formalization targets. The analytic frontier is a reusable theorem for positive Euler products with a controlled central arc and fixed-factor minor-arc suppression. For ProveIt’s transseries development, an algebra of smooth or analytic periodic coefficient germs would encode the phase correctly. One should not label the present contour proof formalized merely because some underlying algebraic operations have Lean counterparts.

11 Conclusion

A033552 admits a concrete, computable all-orders asymptotic description whose most visible unusual feature is a persistent periodic amplitude. The dominant size coordinate is not simply $\log_4 n$: the Catalan polynomial factor shifts the saddle and changes both the logarithmic power and the inverse growth. Keeping that shift, the exact Barnes constant, the local lattice correction, and the Gaussian coefficient corrections in one calculation yields (2.6) and its inverse.

The main analytic issues have explicit proofs: noncentral coefficient arcs are negligible to every fixed inverse-logarithmic order; local lattice Taylor expansions have uniform remainders; saddle reversion has a derivative bounded away from zero; and integer threshold errors are separated from analytic inverse errors. The largest-index theorem gives a probabilistic consequence of the same local phase. What remains is not a missing leading argument, but sharper certification, higher explicit coefficients, secondary exponential sectors, wider statistics, and a systematic formalization.

A A compact audit of the algebra

This appendix records the dependency chain for the constants most likely to be affected by an indexing or sign error.

Quantity	Formula or origin
Allowed unit part	$c_1 = 1$ only; no second c_0 factor
Catalan logarithmic drift	$\beta = 3/2, b_1 = -9/8, b_2 = 1/2$
Summatory logarithmic coefficient	$b_1 - \beta/2 = -15/8$
Barnes constant	$K = -\frac{3}{4} \log(2\pi) - \log G(3/2)$
Bulk $1/m$ coefficient	$27/16$ before the phase correction
First local phase correction	$-(3/2)\partial_\alpha \Psi_\alpha$
Saddle mean correction	$q_0 = -1/2 + \Psi'_\alpha/\alpha$
Logarithmic power of μ	$3\beta/2 - b_1 - 1/2 = 23/8$
Normalized constant	$C = -K - c - \frac{1}{2} \log(2\pi)$
First Gaussian log correction	$-1/(12m)$

The cancellation producing D_α is the identity

$$\begin{aligned} \alpha b + (-A + \Psi'_\alpha + q'_0)a + P_1 + q_1 - b_1 - \frac{q_0}{2} - \frac{1 + q'_0}{2\alpha} - \frac{1}{12} \\ = \frac{1}{6} - \frac{1}{2\alpha} - \frac{3}{2}\partial_\alpha \Psi_\alpha + \frac{17}{8\alpha}\Psi'_\alpha - \frac{(\Psi'_\alpha)^2 + \Psi''_\alpha}{2\alpha^2}, \end{aligned}$$

with a, b, q_0, q_1 as in (5.3) and (5.9)–(5.10). The supplied symbolic audit treats the phase and its derivatives as independent symbols and verifies this rational identity exactly. This check isolates coefficient algebra from the separate analytic justification of the asymptotic operations.

B Provenance and scope of the search

The external check was performed on October 1, 2026. It included the OEIS entry and linked survey, searches for `A033552 asymptotic` and `partitions into Catalan numbers asymptotic`, the classical periodicity reference, and the ProveIt transseries documentation and repository code search. The phrase “Catalan partitions” alone is ambiguous in the literature and was not treated as identifying this sequence without checking the definition.

No claim of an exhaustive MathSciNet, zbMATH, or journal-archive search is made. The absence of a posted OEIS asymptotic is an observable entry-level gap, not evidence that no general theorem could already imply a leading term. Accordingly the main theorems are stated on their mathematical content, and the article does not claim a resolved named conjecture or verified historical priority. No repository file or OEIS entry was edited as part of this work.

C Provenance of the collection edition

[Write note, batch 7301, 1 October 2026.] This appendix was added in the write.

Source. The report is built from one manuscript, number 39 of batch 7301 of ProveIt’s incoming-reports intake: the archive `Catalan_Partitions_Research.zip` (17 files, a 21-page PDF), which arrived in commit `f8c3a392a` and was placed in the collection by commit `9df4ba51a`. Its pinned repository commit is `e6190a94552f119e9a2140e5fea8450a045a0837` (1 October 2026); the delivered `SOURCE_NOTES.md` also names `1f1981f682b2878bde51a6ad40c22777f362fc05` and wrongly calls it a tree (see the note in [section 1](#)). The title-page author line is kept as delivered; the delivered PDF metadata read “Research report prepared with ChatGPT”, and the write replaced that field by the collection’s “AI-assisted research report prepared for Vladimir Reshetnikov”. The report is AI-assisted and unrefereed.

What the write changed. Every one of the 93 delivered labels received the prefix `ctp:`, and every reference was updated with it; this appendix adds one label. The write added the [Write note] paragraphs (after the principal-formula box, in [section 1](#) twice, after the proof of [lemma 4.1](#), in [section 9](#) and here), the bibliography entries [9] and [10], the `\writenote` macro, the PDF author field above, `cleveref` type hints (`\label[lemma]`, `\label[corollary]`, `\label[proposition]`) on the five labels of lemmas, corollaries and propositions, and `\crefalias{section}{appendix}` after `\appendix`. These environments share the theorem counter, and the delivered PDF printed every reference to them as “theorem”; only those printed names changed, no label name or number. No delivered sentence, statement, proof or table was changed or removed.

Delivered names and shipped paths. `article.tex`, `README.md` (rewritten as the collection’s report guide) and `SOURCE_NOTES.md` are at the report root; `code/` keeps its three Python files, and `build.sh` moved there from the package root; `data/` keeps its seven files, and `requirements.txt` moved there from the package root. All but the article and the README are byte-identical to the delivery. Not shipped: the delivered `article.pdf` (replaced by a build of this edition) and the checksum ledger `SHA256SUMS.txt` (verified 16/16 at intake, then retired). `data/build_and_quality_checks.json` describes the delivered 21-page PDF.

Relation to repository material. No repository report or formal development treats A033552. The Mellin computation of [lemma 4.1](#) reappears, for a different product, in the Fabius-tree compilation cited in the note after it [10]. The batch’s other partition reports, on square-root-restricted partitions (A097356) and on distinct-partition norms (A022629), study different products in different saddle regimes and share only the standard saddle, Edgeworth and Lambert- W toolkit; no proposition is proved in two of them.

References

- [1] The OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, A033552, “Number of partitions into Catalan numbers.” Entry inspected October 1, 2026. <https://oeis.org/A033552>.

- [2] I. Pak, *Complexity problems in enumerative combinatorics*, arXiv:1803.06636v2, 2018, especially §2.5. <https://arxiv.org/abs/1803.06636>.
- [3] P. Erdős and B. Richmond, “Concerning periodicity in the asymptotic behaviour of partition functions,” *Journal of the Australian Mathematical Society, Series A* **21** (1976), 447–456. DOI: [10.1017/S1446788700019285](https://doi.org/10.1017/S1446788700019285). Author archive: https://www.renyi.hu/~p_erdos/1976-07.pdf.
- [4] P. Flajolet, X. Gourdon, and P. Dumas, “Mellin transforms and asymptotics: Harmonic sums,” *Theoretical Computer Science* **144** (1995), 3–58. <https://specfun.inria.fr/dumas/Publications/FlGoDu95.pdf>.
- [5] NIST Digital Library of Mathematical Functions, §5.17, “Barnes’ G -function (double gamma function).” <https://dlmf.nist.gov/5.17>. Accessed October 1, 2026.
- [6] NIST Digital Library of Mathematical Functions, §25.10(i), “Zeros: Distribution.” <https://dlmf.nist.gov/25.10.i>. Accessed October 1, 2026.
- [7] V. Reshetnikov, *ProveIt*, repository snapshot e6190a94552f119e9a2140e5fea8450a045a0837; Analysis/Transseries/README.md and Analysis/Transseries/docs/series-and-transseries/README.md. <https://github.com/VladimirReshetnikov/ProveIt/tree/e6190a94552f119e9a2140e5fea8450a045a0837/Analysis/Transseries>.
- [8] The mpmath developers, *Zeta functions, L -series and polylogarithms*, mpmath documentation, including Euler–Maclaurin evaluation options. <https://mpmath.org/doc/1.1.0/functions/zeta.html>. The accompanying numerical run used mpmath 1.3.0; see the recorded version in data/verification.json.
- [9] [Added in the write.] N. G. de Bruijn, “On Mahler’s partition problem,” *Indagationes Mathematicae* **10** (1948), 210–220.
- [10] [Added in the write.] ProveIt, Fabius-tree compilation *Geometric Uniform Convolutions and New Frontiers*, Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/frontier-compilations/Geometric_Uniform_Convolutions_and_New_Frontiers/fabius-frontier-report-H.tex, section “Mellin analysis and the hidden log-periodic phase” (Lemma “Mellin kernel”).